

Chapter 1

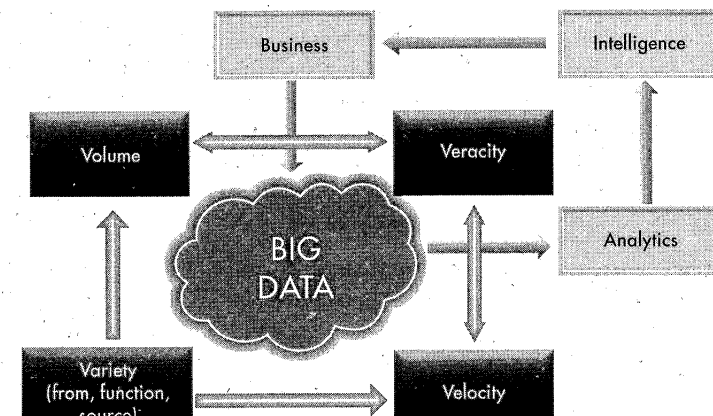
Wholeness of Big Data

Learning Objectives

- Understand Big Data and its powerful business benefits
- Learn the differences between Big Data and Conventional Data
- Learn the 4 Vs of Big Data – Volume, Velocity, Variety, and Veracity
- Discover three types of business applications of Big Data
- Learn valuable tips on how to manage Big Data
- Conceptualize the Big Data ecosystem and appreciate its key components
- Appreciate major technological challenges in managing Big Data
- Relate to the key technology solutions for addressing those challenges

INTRODUCTION

Big Data is an all-inclusive term that refers to extremely large, very fast, highly diverse, and complex data that cannot be managed with traditional data management tools.



Ideally, Big Data includes all kinds of data, which helps deliver the right information, to the right person, in the right quantity, at the right time, to help make the right decisions. Big Data can be harnessed by developing infinitely scalable, flexible, and evolutionary data architectures, coupled with the use of cost-effective computing machines. The infinite potential knowledge embedded within this Big Data cosmic computer would help connect with and enjoy the support of all the laws of nature.

This book will provide a complete overview of Big Data for executives and data scientists. This chapter will cover the key challenges and benefits of Big Data, and the essential tools and technologies available for organizing and manipulating Big Data.

1.1 UNDERSTANDING BIG DATA

Big Data can be examined at two levels (Figure 1.1). At a fundamental level, it is just another collection of data that can be analyzed and utilized for the benefit of the business. On another level, it is a special kind of data that poses unique challenges and offers unique benefits. This book will focus on the latter level.

At business level, data generated by business operations, can be analyzed to generate insights that can help the business make better decisions. This makes the business grow bigger, and generate even more data, so that the cycle continues. This is represented by the cycle on the top-right of Figure 1.1. This aspect is discussed in Chapter 10, 'A Primer on Data Analytics'.

On the other level, Big Data is different from traditional data in every way, i.e. space, time, and function. Quantitatively, Big Data is 1000 times more than the traditional data and of the data generation and transmission speed is also of the same order. The forms and functions of Big Data are 10 times more diverse: from numbers to text, pictures, audio, videos, web logs, machine data, and more. There are many more sources of data, from individuals to organizations to governments, using a range of devices from mobile phones to computers to industrial machines. Not all Big Data will be of equal quality and value, as represented by the cycle on the bottom left of Figure 1.1. This aspect of Big Data, and its new technologies, is the focus of this book.

Big Data is mostly, over 90 per cent, unstructured data. Every type of Big Data is structured differently, and should be dealt in with an appropriate way. There are huge opportunities for technology providers to innovate and manage the entire life cycle of Big Data; to generate, gather, store, organize, analyze, and visualize this data.

Caselet

IBM Watson: A Big Data System

IBM created the Watson system as a way of pushing the boundaries of Artificial Intelligence and natural language understanding technologies. Watson beat the world champion human players of Jeopardy! (quiz style TV show) in Feb 2011. Watson reads up data about everything on the web including the entire Wikipedia. It digests and absorbs the data based on simple generic rules such as: books have authors; stories have heroes; and drugs treat ailments. A Jeopardy! clue, received in the form of a cryptic phrase, is broken down into many possible potential sub-clues of the correct answer. Each sub-clue is examined to see the likeliness of its answer being the correct answer for the main problem. Watson calculates the confidence level of each possible answer. If the confidence level reaches more than a desirable threshold level, Watson decides to offer the answer to the clue. It manages to do all this in a mere 3 seconds.

Watson is now being applied to many important applications such as diagnosing diseases, especially cancer. Watson can read all the new research published in the medical journals to update its knowledge base. It can diagnose the probability of various diseases, by applying factors such as patient's current symptoms, health history, genetic history, medication records, and other factors to recommend a particular diagnosis. (Source: Smartest machines on Earth: youtube.com/watch?v=TCOhyaw5bwg)



FIGURE 1.2 IBM Watson playing jeopardy

1. What kinds of Big Data knowledge, technologies and skills are required to build a system like Watson? What other resources are needed?
2. Will doctors be able to compete with Watson in diagnosing diseases and prescribing medications? Who else could benefit from a system like Watson?

1.2 CAPTURING BIG DATA

If data were only growing too large, or only moving too fast, or only becoming too diverse, it would have been relatively easy. However, when three Vs (Volume, Velocity, Variety) arrive together in an interactive manner, it creates a perfect storm. While the Volume and Velocity of data drive the major technological concerns and the costs of managing Big Data, these two Vs are themselves being driven by the third V, that is, the Variety of forms and functions and sources of data. The varying veracity and value of data complicate the situation further.

1.2.1 Volume of Data

The quantity of data generated in the world has been relentlessly doubling every 12–18 months. Traditional data is measured in Gigabytes (GB) and Terabytes (TB), but Big Data is measured in Petabytes (PB) and Exabytes (1 Exabyte = 1 Million TB). This data is so huge that it is almost a miracle that one can find any specific thing in it, in a reasonable period of time. Searching the world-wide web was the first true Big Data application. Google perfected the art of this application and developed many of the path-breaking Big Data technologies we see in use today.

The primary reason for the growth of data is the dramatic reduction in the cost of storing data. The costs of storing data have decreased by 30–40 per cent every year. Therefore, there is an incentive to record everything that can be observed. It is called 'datafication' of the world. The costs of computation and communication of data have also been coming down. Another reason for the growth of data is the increase in the number of forms and functions of data. We will discuss more about this in the 'Variety' section.

1.2.2 Velocity of Data

If traditional data is like a lake, Big Data is like a fast-flowing river. Big Data is being generated by billions of devices, and communicated at the speed of the light, through the internet. Ingesting all this data is like drinking from a fire hose. One does not have any control over how fast the data will come. A huge unpredictable data-stream is the new metaphor for thinking about Big Data.

The primary reason for the increased velocity of data is the increase in internet speed. Internet speeds available to homes and offices are now increasing from 10 MB/s to 1 GB/s (100 times faster). More people are getting access to high-speed internet around the world. Another important reason is the increased variety of sources, such as mobile devices, that can generate and communicate data from anywhere, at any time.

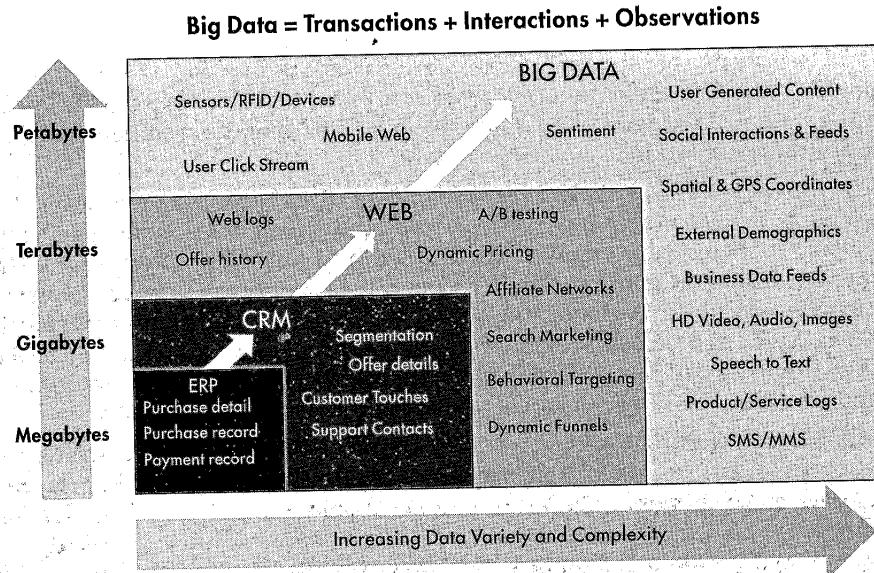
1.2.3 Variety of Data

Big Data is inclusive of all forms of data, for all kinds of functions, from all sources and devices. If traditional data forms such as invoices and ledgers were like a small store, Big Data is the biggest imaginable shopping mall that offers unlimited variety. There are three major kinds of variety of data.

1. The *first* aspect of variety is the **Form** of data. Data types range in variety from numbers to text, graph, map, audio, video, and others, with some being simple and others being very complex. There could be composites of data that include many elements in a single file. For example, text documents have graphs and pictures embedded in them. Video movies can have audio songs embedded in them. Audio and video have different and vastly more complex storage formats than numbers and text. Numbers and text can be more easily analyzed than an audio or video file. How should composite entities be stored and analyzed?
2. The *second* aspect of variety is the **Function** of data. There is data from human conversations, songs and movies, business transaction records, machine operations performance data, new product design data, old archived data, etc. Human communication data would need to be processed very differently from operational performance data, with different expectations and objectives. Big Data technologies could be used to recognize people's faces in pictures; compare voices to identify the speaker; and compare handwritings to identify the writer.
3. The *third* aspect of variety is the **Source** of data. Mobile phone and tablet devices enable a wide series of applications (or apps) to access data and, also, generate data from anytime anywhere. Web access and search logs are another new and huge source of data. Business systems generate massive amount of structured business transactional information. Temperature and pressure sensors on machines, and Radio Frequency (RFID) tags on assets, generate incessant and repetitive data. Broadly speaking, there are three broad types of sources of data: Human-human communications; human-machine communications; and machine-to-machine communications. The sources of Big Data, and their business applications, will be discussed in the next chapter.

1.2.4 Veracity of Data

Veracity relates to the truthfulness, believability, and quality of data. Big Data is messy and there is considerable misinformation and disinformation out there. The reasons for poor quality of data can range from technical error, to human error, to malicious intent.



Source: Contents of above graphic created in partnership with Teradata, Inc.

FIGURE 1.3 Sources of big data (Source: Hortonworks.com)

1. The source of information may not be authoritative. For example, all websites are not equally trustworthy. Any information from whitehouse.gov or from nytimes.com is more likely to be authentic and complete. Wikipedia is useful, but not all pages are equally reliable. In each case, the communicator may have an agenda or a point of view.
2. The data may not be communicated and received correctly because of human or technical failure. Sensors and machines for gathering and communicating data may malfunction and may record and transmit incorrect data. Urgency may require the transmission of the best data available at a point in time. Such data makes reconciliation with later, accurate, records more problematic.
3. The data provided and received may, however, also be intentionally wrong for competitive or security reasons. There could be disinformation and malicious information spread for strategic reasons.

Big Data need to be sifted and organized by quality, for it to be put to any great use.

1.3 BENEFITTING FROM BIG DATA

Data is the new natural resource. Traditionally, data usually belongs to the organization that generates it. There are other types of data, such as social media data, which are freely accessible under an open general license. Organizations can use this data to learn about their consumers, improve their service delivery, and design new products to delight their customers, and to gain a competitive advantage. Data is also being used to design new digital products, such as on-demand entertainment and learning.

Organizations cannot afford to ignore Big Data. They may choose to gather and store this data for later analysis, or to sell it to other organizations, that might benefit from it. They may also legitimately choose to discard parts of their data for privacy or legal reasons. However, organizations that do not learn to engage with Big Data, could find themselves left far behind their competition, landing in the dustbin of history. Innovative and nimble small organizations can use Big Data to quickly scale up and beat larger and more mature organizations.

Big Data applications exist in all industries and aspects of life. There are three major types of Big Data applications: Monitoring and Tracking, Analysis and Insight, and Digital Product Development.

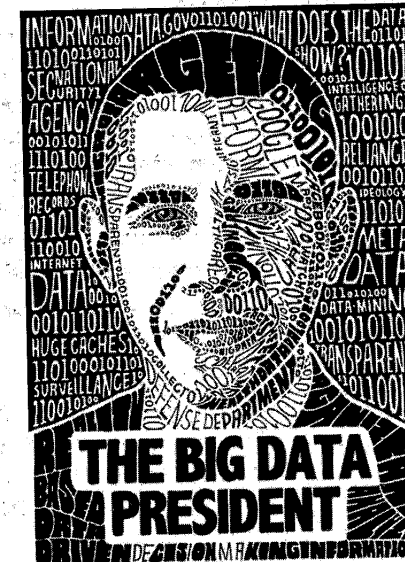


FIGURE 1.4 The first big data president

Monitoring and Tracking Applications Consumer goods producers use monitoring and tracking applications to understand the sentiments and needs of their customers. Industrial organizations use Big Data to track inventory in massive interlinked global supply chains. Factory owners use it to monitor machine performance and do preventive maintenance. Utility companies use it to predict energy consumption, and manage demand and supply. Information Technology companies use it to track website performance and improve its usefulness. Financial organizations use it to project trends better and make more effective and profitable bets, etc.

Analysis and Insight Political organizations use Big Data to micro-target voters and win elections. Police use Big Data to predict and prevent crime. Hospitals use it to better diagnose diseases and make medicine prescriptions. Advertising agencies use it to design more targeted marketing campaigns more quickly. Fashion designers use it to track trends and create more innovative products.

New Product Development Incoming data could be used to design new products such as reality TV entertainment. Stock market feeds could be a digital product. Imagination is the limit on how new such products and services can be developed and delivered at the speed of thought.

1.4 MANAGEMENT OF BIG DATA

Big Data is a new phase representing a digital world. Business and society are not immune from its strong impacts. Many organizations have started initiatives around the use of Big Data. However, most organizations do not necessarily have a good grip on it. Here are some emerging insights into making better use of Big Data.

1. Across all industries, the business case for Big Data is strongly focused on addressing customer-centric objectives. The first focus on deploying Big Data initiatives is to protect and enhance customer relationships and customer experience.
2. Big Data should be used to solve a real pain-point. It should be deployed for specific business objectives of management while avoiding being overwhelmed by the enormity of Big Data.
3. Organizations are beginning their pilot implementations by using existing and newly accessible internal sources of data. It is better to begin with data under one's control and where one has a superior understanding of the data. One should use more diverse data, and not just more data. This would permit tuning into a broader perspective of reality, leading to better insights.

4. Putting humans and data together leads to the most insights. Combining data-based analysis with human intuition and perspectives is better than going just one way. Most organizations lack the advanced analytical capabilities required to get the most value from Big Data. There is a growing awareness of building or hiring those skills and capabilities.
5. The faster one analyzes the data, the more its predictive value. The value of data depreciates rapidly with time. Many kinds of data need to be processed within minutes, or else the immediate competitive advantage is lost.
6. One should not throw away data if no immediate use can be seen for it. Data usually has value beyond what one initially anticipates. Data can add perspective to other data later in a multiplicative manner. One should maintain just one copy of your data, not multiple. This would help avoid confusion and increase efficiency.
7. Big Data is growing exponentially, so one should plan for exponential growth. Storage costs continue to fall, data generation continues to grow, data-based applications continue to grow in capability and functionality.
8. Big Data builds upon a resilient, secure, efficient, flexible, and real-time information processing environment. Big Data is transforming business, just like IT did.

1.5 ORGANIZING BIG DATA

Good organization depends upon the purpose of the organization.

Given huge quantities of data, it would be desirable to organize the data to speed up the search process for finding a specific desired thing in the entire data. The cost of storing and processing the data, too, would be a major driver for the choice of an organizing pattern.

Given fast and variable speed of data, it would be desirable to create a scalable number of ingest points. It will also be desirable to create at least some element of control over the data by maintaining count and averages over time, unique values received, etc.

Given wide variety in form factors, data will need to be stored and analyzed differently. Videos need to be stored separately and used for serving in a streaming mode. Text data may be stored separately, so it can be combined, cleaned, and analyzed for themes and sentiments.

Given different quality of data, various data sources may need to be ranked and prioritized before serving them to the audience. For example, the quality of a webpage and its data may be evaluated using its PageRank value.

1.6 ANALYZING BIG DATA

Big Data can be analyzed in two ways. Big Data can be utilized to visualize a flowing or a static situation. These are called analyzing Big Data in motion AND Big Data at rest. First way is to process the incoming stream of data in real time for quick and effective statistics about the data. The second way is to store and structure batches of data and apply standard analytical techniques on for generating insights. This could then be visualized using real-time dashboards. The nature of processing this huge, diverse, and largely unstructured data, can be limited only by one's imagination.

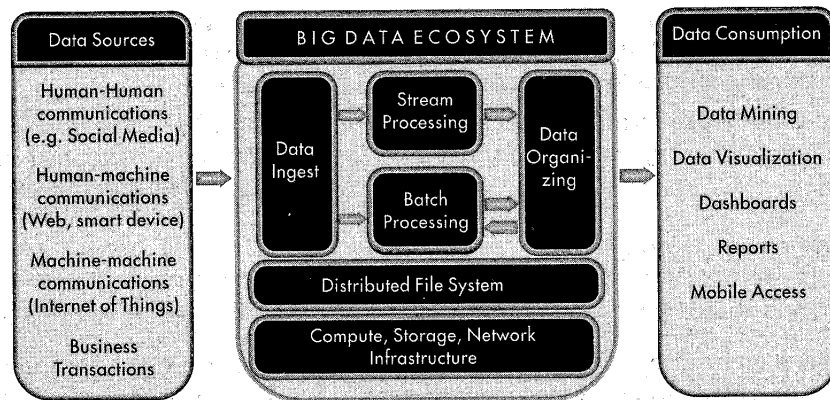


FIGURE 1.5 Big data architecture

A million points of data can be plotted in a graph and offer a view of the density of data. However, plotting a million points on the graph may produce a blurred image which may hide, rather than highlight the distinctions. In such a case, binning the data would help, or selecting the top few frequent categories may deliver greater insights. Streaming data can also be visualized by simple counts and averages over time. For example, below is a dynamically updated chart that shows up-to-date statistics of visitor traffic to the author's blogsite, anilmah.com. The bar shows the number of page views, and the inner darker bar shows the number of unique visitors. The dashboard could show the view by days, weeks, or years also.

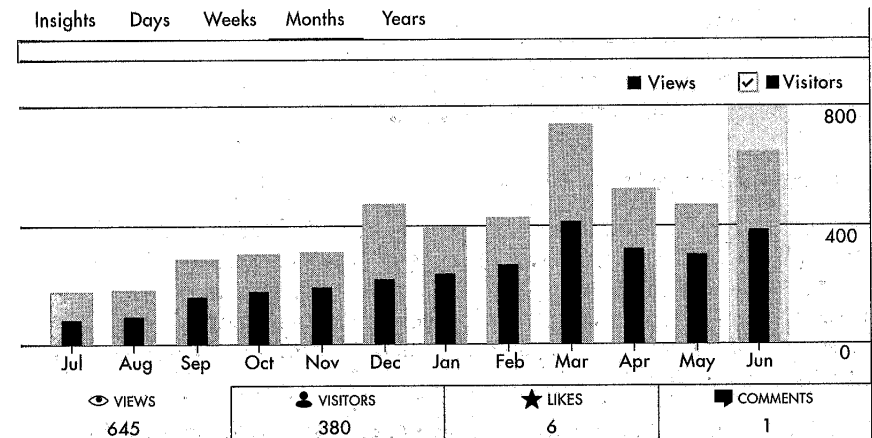


FIGURE 1.6 Real-time dashboard for website performance for the author's blog

Text Data could be combined, filtered, cleaned, thematically analyzed, and visualized in a wordcloud. Here is wordcloud from a recent stream of tweets (i.e. Twitter messages) from 2016 US Presidential candidates Hillary Clinton and Donald Trump (Figure 1.7). The larger size of words implies greater frequency of occurrence in the tweets. This can help understand the major topics of discussion between the two.

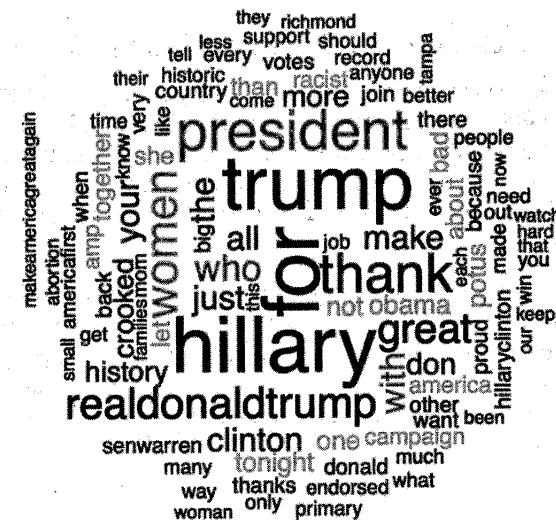


FIGURE 1.7 A wordcloud of Hillary Clinton's and Donald Trump's tweets

1.7 TECHNOLOGY CHALLENGES FOR BIG DATA

There are four major technological challenges in managing Big Data. There are four matching layers of technologies to overcome those challenges.

1.7.1 Storing Huge Volumes

The first challenge relates to storing huge quantities of data. No computing machine is big enough to store the relentlessly growing quantity of data. Therefore, data must be stored in several smaller inexpensive machines. However, with a large number of machines, there is the inevitable challenge of machine failure. Each of these commodity machines will fail at some point or another. Failure of a machine could entail a loss of data stored on it.

The first layer of Big Data technology thus helps store huge volumes of data, at an affordable cost, while avoiding the risk of data loss. It distributes data across a large cluster of inexpensive commodity machines. It ensures that every piece of data is stored on multiple machines to guarantee that at least one copy is always available.

Hadoop is the most well-known clustering technology for Big Data. Its data storage pattern is called Hadoop Distributed File System (HDFS). This system is built on the patterns of Google's Big File systems, designed to store billions of pages, and sort them to answer user search queries.

1.7.2 Ingesting Streams at an Extremely Fast Pace

The second challenge relates to the Velocity of data, i.e. handling torrential streams of data. Some of the data streams may be too large to store, but must still be monitored. The solution lies in creating special scalable ingesting systems that can open an unlimited number of channels for receiving data. These systems can hold data in queues, from which business applications can read and process data at their own pace and convenience.

The second layer of Big Data technology manages this velocity challenge. It uses a special stream-processing engine, where all incoming data is fed into a central queueing system. From there, a fork-shaped system sends data to batch storage as well as to stream processing directions. The stream processing engine can do its work while the batch processing does its work. Apache Spark is the most popular system for streaming applications.

1.7.3 Handling a Variety of Forms and Functions of Data

The third challenge relates to the structuring and access of all varieties of data that comprise Big Data. Storing them in traditional flat or relational structures would be too impractical, wasteful, and slow. Accessing and analyzing them requires different capabilities.

The third layer of Big Data technology solves this problem by storing the data in non-relational systems that relax many of the stringent conditions of the relational model. These are called NoSQL (Not Only SQL) databases. These databases are optimized for certain tasks such as query processing, or graph processing, document processing, etc.

HBase and Cassandra are two of the better known NoSQL databases systems. HBase, for example, stores each data element separately along with its key identifying information. This is called a key-value pair format. Cassandra stores data in a columnar format. There are many other variants of NoSQL databases. NoSQL languages, such as Pig and Hive, are used to access this data.

1.7.4 Processing Data at Huge Speeds

The fourth challenge relates to moving large amounts of data from storage to the processor, as this would consume enormous network capacity and choke the network. The alternative and innovative mode would be to do just the opposite, i.e. to move the processing to where the data is stored.

The fourth layer of Big Data technology avoids the choking of the network. It distributes the task logic throughout the cluster of machines where the data is stored. Those machines work, in parallel, on the data assigned to them, respectively. A follow-up process consolidates the outputs of all the small tasks and delivers the final results.

MapReduce, also invented by Google, is the best-known technology for parallel processing of distributed Big Data. Table 1.1 lists technological challenges and solutions for Big Data.

Table 1.1

Technological challenges and solutions for big data

Challenge	Description	Solution	Technology
Volume	Avoid risk of data loss from machine failure in clusters of commodity machines	Replicate segments of data in multiple machines; master node keeps track of segment location	HDFS

(Contd.)

(Contd.)

Challenge	Description	Solution	Technology
Volume & Velocity	Avoid choking of network bandwidth by moving large volumes of data	Move processing logic to where the data is stored; manage using parallel processing algorithms	Map-Reduce
Variety	Efficient storage of large and small data objects	Columnar databases using key-pair values format	HBase, Cassandra
Velocity	Monitoring streams too large to store	Fork-shaped architecture to process data as stream and as batch	Spark

Once these major technological challenges arising from the 4 Vs of data are met, all traditional analytical and presentation tools can be applied to Big Data. There are many additional supportive technologies to make the task of managing Big Data easier. For example, there are technologies to monitor the resource usage and load balancing of the machines in the cluster.

1.8 CONCLUSION

Big Data is a major social and technological phenomenon that impacts everyone. It also provides an opportunity to create new ways of knowing and working. Big Data is extremely large, complex, fast, and not always clean as it is data that comes from many sources such as people, web, and machine communications. Big Data needs to be gathered, organized and processed in a cost-effective way that manages the volume, velocity, variety, and veracity of Big Data. Hadoop, MapReduce, NoSQL, and Spark systems are popular technological platforms for this purpose.

To summarize, here is a list of the many differences between traditional and Big Data (Table 1.2).

Table 1.2
Comparing big data with traditional data

Feature	Traditional Data	Big Data
Representative Structure	Lake/Pool	Flowing Stream/river
Primary Purpose	Manage business activities	Communicate, Monitor
Source of data	Business transactions, documents	Social media, Web access logs, machine generated
Volume of data	Gigabytes, Terabytes	Petabytes, Exabytes
Velocity of data	Ingest level is controlled	Real-time unpredictable ingest
Variety of data	Alphanumeric	Audio, Video, Graphs, Text

(Contd.)

(Contd.)

Feature	Traditional Data	Big Data
Veracity of data	Clean, more trustworthy	Varies depending on source
Structure of data	Well-Structured	Semi- or Un-structured
Physical Storage of Data	In a Storage Area Network	Distributed clusters of commodity computers
Database organization	Relational databases	NoSQL databases
Data Access	SQL	NoSQL such as Pig
Data Manipulation	Conventional data processing	Parallel processing
Data Visualization	Variety of tools	Dynamic dashboards with simple measures
Database Tools	Commercial systems	Open-source - Hadoop, Spark
Total Cost of System	Medium to High	high

1.9 ORGANIZATION OF THE REST OF THE BOOK

This book will cover applications, architectures, and the essential Big Data technologies. The rest of the book is organized as follows.

Section 1 will discuss sources, applications, and architectural topics. Chapter 2 will discuss a few compelling business applications of Big Data, based on understanding the different sources and formats of data. Chapter 3 will cover some examples of Big Data architectures used by leading organizations.

Section 2 will discuss expand on the Big Data Ecosystem (Figure 1.5). It will have six chapters discussing the six major technology elements identified in the ecosystem. Chapter 4 will discuss Hadoop ecosystem and how its Distributed File system (HDFS) works. Chapter 5 will discuss MapReduce parallel processing algorithm and how it helps accomplish big tasks quickly. An introduction to Hive and Pig programming languages will also be included. Chapter 6 will discuss NoSQL databases to learn how to structure Big Data into four major types of databases. In particular, HBase and Cassandra databases will be covered in great detail. Chapter 7 will cover streaming data, and the systems for ingesting and processing this data. This chapter will cover Spark, an integrated, in-memory processing toolset to manage Big Data. Chapter 8 will cover Data ingest system, using Apache Kafka messaging system. Finally, Chapter 9 will be a primer on Cloud Computing technologies used for renting storage and compute capacity at third party networks at a fraction of the cost of in-house IT infrastructure.

Section 3 will include applications, primers, and tutorials. Chapter 10 will present a case study on "web log analyzer", a popular and generic application that ingests a log of a large number of web request entries every day and can create summary

and exception reports. Chapter 11 will be a primer on data analytics technologies for analyzing data: A full treatment can be found in my book, *Data Analytics*. Chapter 12 will be a primer on Big Data languages, Apache Hive and Pig. Appendix A will be a tutorial on installing Hadoop cluster on Amazon EC2 cloud. Appendix B will be a tutorial on installing and using Spark.



Review Questions

1. What is Big Data? Why should anyone care?
2. Describe the 4V model of Big Data.
3. What are the major technological challenges in managing Big Data?
4. What are the technologies available to manage Big Data?
5. What kind of analyses can be done on Big Data?
6. Watch Cloudera CEO present the evolution of Hadoop at <https://www.youtube.com/watch?v=S9xnYBVqLws>. Why did people not pay attention to Hadoop and MapReduce when it was introduced? What implications does it have to emerging technologies?



True/False Questions

1. Big Data is an umbrella term for a collection of data sets so large and complex that they do not fit into a single file system.
2. 'Datafication' means that 'increasing' details about events are being recorded and stored.
3. Data only belongs to the user or the organization that generates it.
4. Big Data is so precious that multiple copies of data should be maintained.
5. Big Data includes approximately equal amounts of structured data and unstructured data.
6. Social media interactions are a form of big data.
7. Big Data is growing steadily at an exponential rate.
8. Big Data at rest means to process the incoming stream of data in real time for quick and effective statistics about the data.
9. The major causes of variety of data are the source, form, and function of data.

10. Transactions + Interactions + Observations = Traditional Data.
11. The primary cause of poor quality of data is because of human error.
12. There are five different categories of Big Data applications.
13. Big Data should be used for solving technology-centric applications.
14. Visualizing Big Data presents no special challenges.
15. Storing huge volumes of data is solved using MapReduce.
16. Processing massive data in parallel is solved using NoSQL databases.
17. Processing massive data in real-time streaming mode is done through Apache Spark.
18. Managing huge variety of data is done through NoSQL databases.
19. Hadoop is a proprietary Big data technology.
20. Hadoop Distributed File System (HDFS) was invented by Google.

Liberty Stores Case Exercise: Step B1

Liberty Stores Inc. is a specialized global retail chain that sells organic food, organic clothing, wellness products, and education products to enlightened LOHAS (Lifestyles of the Healthy and Sustainable) citizens worldwide. The company is 20 years old, and is growing rapidly. It now operates in 5 continents, 50 countries, 150 cities, and has 500 stores. It sells 20000 products and has 10000 employees. The company has revenues of over \$5 billion and has a profit of about 5 per cent of its revenue. The company pays special attention to the conditions under which the products are grown and produced. It donates about one-fifth (20 per cent) from its pre-tax profits from global local charitable causes.

1. Suggest a comprehensive Big Data strategy for the CEO of the company.
2. How can a Big Data system such as IBM Watson help this company?